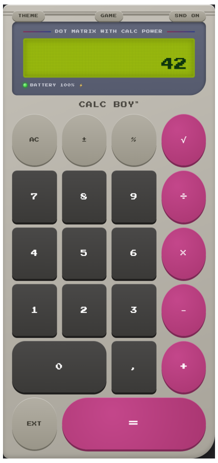
CALC BOY

User Manual



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Introduction

CALC BOY is a handheld-console-style calculator. This manual teaches the everyday workflows, explains when to use each page, and shows the results you should expect.

Tip: The display is also interactive. Tap it for history; long-press it to copy the current result.

Installation

- Open the live link in Safari or Chrome.
- Use Share or the browser menu and choose Add to Home Screen.
- Launch CALC BOY from the home screen. Installed mode opens the app fullscreen.

Offline Mode

After the first online launch, the browser keeps the app available for later. CALC BOY can then open without a connection as long as the site's stored data remains on the device.

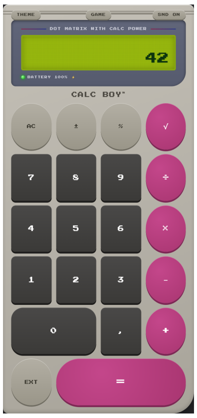
First calculation

Example: 12 + 30 = shows 42 and adds the expression to history.

History, copy and share

The last ten successful calculations are saved. The history view also shows sum and average. Text and PNG export are available when the browser supports sharing or clipboard APIs.

BASIC



What this page does	Standard calculator for everyday arithmetic, signs, square roots and smart percent calculations.
How to use it	Enter the first number, choose an operator, enter the second number, then press =. Use AC to clear the current calculation. After ERROR, the next digit starts a new entry. Tap the LCD to open history; long-press the LCD to copy the displayed result.
Important details	The percent key is context-aware: with + or - it calculates a percentage of the first operand.

Examples

Example	Steps
Smart percent	100 + 10 % = 110; 100 - 10 % = 90
Square root	144 sqrt -> 12

Button reference

Key	Function	Key	Function
AC	Clears entry and active operation; in RPN it also clears the stack.	*	Multiplies two values.
+/-	Changes the sign of the displayed number.	1	Digit input.
%	Smart percent. Example: 100 + 10 % becomes 110.	2	Digit input.
sqrt	Calculates the square root; negative values show ERROR.	3	Digit input.
7	Digit input.	-	Subtracts the second value from the first.
8	Digit input.	0	Digit input.
9	Digit input.	,	Adds the decimal separator.
/	Divides the first value by the second; division by zero shows ERROR.	+	Adds two values.
4	Digit input.	EXT	Opens the extended scientific page.
5	Digit input.	=	Evaluates the current calculation; in RPN it pushes the value to the stack.
6	Digit input.		

EXT

What this page does	Scientific and memory page with trigonometry, logarithms, powers, roots, factorial, pi and e.
How to use it	For direct functions such as sin, log, x^2 or 1/x, enter a value and press the function key. For x^y, enter the base, press x^y, enter the exponent, then press =. Use DEG/RAD before trigonometry. DEG is saved and is the default unless changed.
Important details	Factorial works only for whole numbers from 0 to 170.

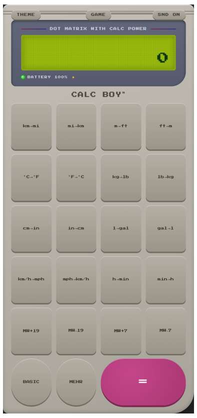
Examples

Example	Steps
DEG/RAD	DEG: 30 sin -> 0.5; RAD: pi sin -> approximately 0
Memory	42 M+, AC, MR -> 42; 10 M- leaves 32 in memory
Powers	2 x^y 8 = -> 256

Button reference

Key	Function	Key	Function
MC	Clears calculator memory.	x^2	Squares the displayed value.
MR	Recalls the saved memory value to the display.	x^3	Cubes the displayed value.
M+	Adds the displayed value to memory.	x^y	Binary power: enter base, press x^y, enter exponent, press =.
M-	Subtracts the displayed value from memory.	cbrt	Calculates the cube root.
sin	Sine of the displayed value, using DEG or RAD mode.	1/x	Calculates the reciprocal; 0 shows ERROR.
cos	Cosine of the displayed value, using DEG or RAD mode.	n!	Calculates factorial for whole numbers from 0 to 170.
tan	Tangent of the displayed value, using DEG or RAD mode.	pi	Inserts pi.
DEG/RAD	Toggles saved angle mode for trigonometry.	e	Inserts Euler's number e.
log	Base-10 logarithm of the displayed value.	BASIC	Returns to the standard calculator page.
ln	Natural logarithm of the displayed value.	MEHR	Cycles EXT -> CONV -> FIN -> PRG -> PLOT -> FORM.
10^x	Calculates ten to the power of the displayed value.	=	Evaluates the current calculation; in RPN it pushes the value to the stack.
e^x	Calculates e to the power of the displayed value.		

CONV



What this page does	One-tap converters for distance, temperature, mass, volume, speed, time and German VAT.
How to use it	Enter the source value and press the desired conversion key. Use the matching reverse key if you need to convert back. VAT keys treat the entered value as net or gross according to the key label.
Important details	US gallons are used for litre/gallon conversion.

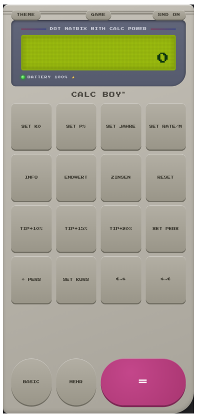
Examples

Example	Steps
VAT	100 MW+19 -> 119; 119 MW-19 -> 100
Temperature	20 C->F -> 68; 68 F->C -> 20
Speed	100 km/h->mph -> 62.137119224

Button reference

Key	Function	Key	Function
km->mi	Kilometres to miles.	km/h->mph	Kilometres per hour to miles per hour.
mi->km	Miles to kilometres.	mph->km/h	Miles per hour to kilometres per hour.
m->ft	Metres to feet.	h->min	Hours to minutes.
ft->m	Feet to metres.	min->h	Minutes to hours.
C->F	Celsius to Fahrenheit.	MW+19	Adds 19 percent German VAT to a net amount.
F->C	Fahrenheit to Celsius.	MW-19	Removes 19 percent German VAT from a gross amount.
kg->lb	Kilograms to pounds.	MW+7	Adds 7 percent German VAT to a net amount.
lb->kg	Pounds to kilograms.	MW-7	Removes 7 percent German VAT from a gross amount.
cm->in	Centimetres to inches.	BASIC	Returns to the standard calculator page.
in->cm	Inches to centimetres.	MEHR	Cycles EXT -> CONV -> FIN -> PRG -> PLOT -> FORM.
l->gal	Litres to US gallons.	=	Evaluates the current calculation; in RPN it pushes the value to the stack.
gal->l	US gallons to litres.		

FIN



What this page does	Finance helpers for compound interest with monthly savings, interest amount, tips, bill splitting and manual currency conversion.
How to use it	Set K0, P%, years and monthly rate with SET keys. ENDWERT calculates the future value. ZINSEN shows only the earned interest: future value minus start capital and deposits. Set persons before / PERS. Set exchange rate before EUR->$ or $->EUR.
Important details	Default values are K0 1000, P 3, years 10, monthly rate 50, persons 2 and exchange rate 1.08.

Examples

Example	Steps
Compound interest	SET K0=1000, SET P%=3, SET JAHRE=10, SET RATE/M=50, ENDWERT -> 8336.42449101
Interest only	With the same values, ZINSEN -> 1336.42449101
Bill split	SET PERS=4, enter 80, / PERS -> 20
Currency	SET KURS=1.08, 100 EUR->$ -> 108; 108 $->EUR -> 100

Button reference

Key	Function	Key	Function
SET K0	Stores start capital for compound interest.	TIP+20%	Adds 20 percent tip to the displayed amount.
SET P%	Stores yearly interest rate in percent.	SET PERS	Stores number of people for bill splitting.
SET JAHRE	Stores duration in years.	/ PERS	Divides the displayed amount by the stored number of people.
SET RATE/M	Stores the monthly savings amount.	SET KURS	Stores the manual exchange rate.
INFO	Shows stored parameters or variables.	EUR->$	Converts euros to dollars with the stored rate.
ENDWERT	Calculates future value including start capital and monthly payments.	$->EUR	Converts dollars to euros with the stored rate.
ZINSEN	Shows only earned interest.	BASIC	Returns to the standard calculator page.
RESET	Restores finance defaults.	MEHR	Cycles EXT -> CONV -> FIN -> PRG -> PLOT -> FORM.
TIP+10%	Adds 10 percent tip to the displayed amount.	=	Evaluates the current calculation; in RPN it pushes the value to the stack.
TIP+15%	Adds 15 percent tip to the displayed amount.		

PRG

What this page does	Programmer functions, integer base display, bitwise operators, random number and RPN mode.
How to use it	HEX, BIN and OCT show the integer part of the display in the status line. AND, OR, XOR, MOD, << and >> are binary operators: enter first value, operator, second value, =. RPN changes = into push: enter a value, press =, enter the next value, then press an operator.
Important details	Only the integer part of the number is used for bit operations and base display.

Examples

Example	Steps
Base display	255 ->HEX shows FF; ->BIN shows 11111111
Binary operations	5 AND 3 = 1; 5 OR 2 = 7; 9 MOD 4 = 1
RPN	RPN on: 12 =, 3 + -> 15; AC clears the stack

Button reference

Key	Function	Key	Function
->HEX	Shows the integer part in hexadecimal in the status line.	ABS	Absolute value.
->BIN	Shows the integer part in binary in the status line.	INT	Integer part without decimals.
->OCT	Shows the integer part in octal in the status line.	SGN	Sign of the number: -1, 0 or 1.
NOT	Bitwise NOT of the integer part.	ZUFALL	Random integer from 1 to 100.
AND	Bitwise AND for two integer operands.	pi	Inserts pi.
OR	Bitwise OR for two integer operands.	e	Inserts Euler's number e.
XOR	Bitwise XOR for two integer operands.	RPN	Toggles Reverse Polish Notation mode.
MOD	Integer remainder; modulo by 0 shows ERROR.	BASIC	Returns to the standard calculator page.
<<	Shifts the integer part left.	MEHR	Cycles EXT -> CONV -> FIN -> PRG -> PLOT -> FORM.
>>	Shifts the integer part right.	=	Evaluates the current calculation; in RPN it pushes the value to the stack.

PLOT

What this page does	Draws 20 built-in function graphs as pixel plots on the LCD.
How to use it	Open PLOT and press a function key such as sin x, x^2 or GAUSS. The graph replaces the LCD temporarily and scales itself automatically. Press any other key or tap the display to close the graph.
Important details	Asymptotes such as tan x or 1/x are clipped so the plot stays readable.

Examples

Example	Steps
Graph	sin x draws y = sin x; any non-plot key closes it
Compare	x^2 and x^3 use different automatic scales

Button reference

Key	Function	Key	Function
sin x	Draws y = sin x.	sinh	Draws hyperbolic sine.
cos x	Draws y = cos x.	cosh	Draws hyperbolic cosine.
tan x	Draws y = tan x with clipping near asymptotes.	tanh	Draws hyperbolic tangent.
x^2	Draws y = x^2.	x^4	Draws y = x^4.
x^3	Draws y = x^3.	GAUSS	Draws the Gaussian bell curve e^(-x^2).
sqrt x	Draws y = square root of x.	FLOOR	Draws the floor step function.
log x	Draws y = log10 x.	sinc x	Draws sin x / x, with value 1 at x = 0.
1/x	Calculates the reciprocal; 0 shows ERROR.	x*sin x	Draws y = x times sin x.
e^x	Draws y = e^x.	BASIC	Returns to the standard calculator page.
ln x	Draws y = ln x.	MEHR	Cycles EXT -> CONV -> FIN -> PRG -> PLOT -> FORM.
|x|	Draws the absolute-value graph.	=	Evaluates the current calculation; in RPN it pushes the value to the stack.
2^x	Draws y = 2^x.		

FORM

What this page does	Formula assistant based on three stored variables A, B and C.
How to use it	Enter a number and press SET A, SET B or SET C to store it. Press INFO to check stored values. CLR ABC resets all variables to 0. Press a formula key; the result appears on the display and is added to history.
Important details	Formula keys use fixed roles: for example BMI uses A as kg and B as metres; speed uses A as kilometres and B as hours.

Examples

Example	Steps
Percent	SET A=20, SET B=150, % VON -> 30
Pythagoras	SET A=3, SET B=4, PYTH -> 5
BMI	SET A=80, SET B=1.8, BMI -> 24.6913580247
Net/gross	SET A=119, NET19 -> 100; SET A=100, BRU19 -> 119

Button reference

Key	Function	Key	Function
SET A	Stores the displayed value in variable A.	BMI	BMI with A as kg and B as metres.
SET B	Stores the displayed value in variable B.	AVG ABC	Average of A, B and C.
SET C	Stores the displayed value in variable C.	CLR ABC	Resets A, B and C to 0.
INFO	Shows stored parameters or variables.	KM/H	Speed: A kilometres divided by B hours.
% VON	Calculates A percent of B.	NET19	Net amount from a gross amount with 19 percent VAT.
DREISATZ	Calculates B times C divided by A.	BRU19	Gross amount from a net amount with 19 percent VAT.
KREIS A	Circle area with radius A.	BASIC	Returns to the standard calculator page.
KREIS U	Circle circumference with radius A.	MEHR	Cycles EXT -> CONV -> FIN -> PRG -> PLOT -> FORM.
PYTH	Pythagoras: square root of A^2 + B^2.	=	Evaluates the current calculation; in RPN it pushes the value to the stack.
OHM	Ohm helper: A times B.		

Plotting

A plot replaces the LCD with a pixel graph. Press any non-plot key or tap the display to close it.

Games

Math Attack asks arithmetic questions for 30 seconds. Snake uses the number pad 2/4/6/8 or arrow keys and stores its own high score.

Landscape mode

In landscape, the scientific shortcut column is visible beside BASIC, giving quick access to sin, cos, tan, log, ln, powers, pi and e.

Keyboard

Key	Action
0-9	digits
, or .	decimal separator
+ - * /	basic operators
Enter or =	equals
Escape	AC
%	percent
r or w	square root
arrow keys	Snake direction and secret-code input

Storage and Privacy

All settings stay on your device. CALC BOY does not use analytics, does not load external fonts, and does not fetch live exchange rates.

theme, sound, angle mode, history, memory, finance parameters, exchange rate, persons, RPN mode, RPN stack, formula variables, game high scores, Virtual Boy unlock.

Limitations

Offline use requires a first launch from the website. Currency conversion uses the manually stored rate and does not fetch live rates. Programmer base conversion appears in the status line and does not replace the main display. Sharing, clipboard, vibration and battery display depend on browser support. The in-app interface is German.

Troubleshooting

If an old version appears after update, close and reopen the app twice. If sound does not play, tap once inside the app and check SND ON. If sharing is unavailable, use clipboard copy or browser download fallback. To reset all data, delete website data for this domain.

Version Information

CALC BOY 3.1.0 - calcboy-v3.1.0 - GPL-3.0-or-later